

Graph-driven geospatial analysis



ceci n'est pas une conférence

Not just the science of where, but the science of how things are connected across regions and relationships.

Brussels, 25 September 2025

Benjamin DESWYSEN



Benjamin Deswysen

Geodata specialist @GeoPostcodes



Polleur (Liège, Belgium)



ULiege, UNamur



Geo data science & engineering



OSM occasional contributor



© Taïs Grippa - 2023

Graph-driven geospatial analysis

01

What is a graph?
Quick introduction

03

Routing.
pgrouting

02

Handling topology.
Postgis topology

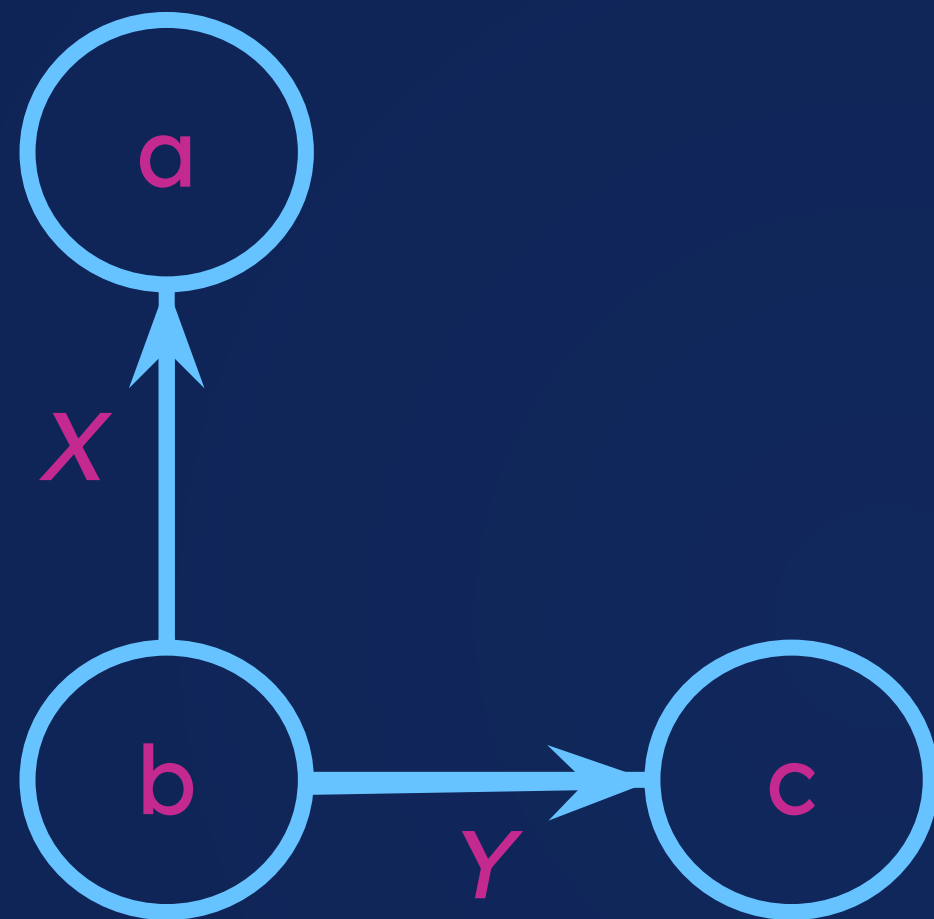
04

Knowledge graphs.
Neo4j

01

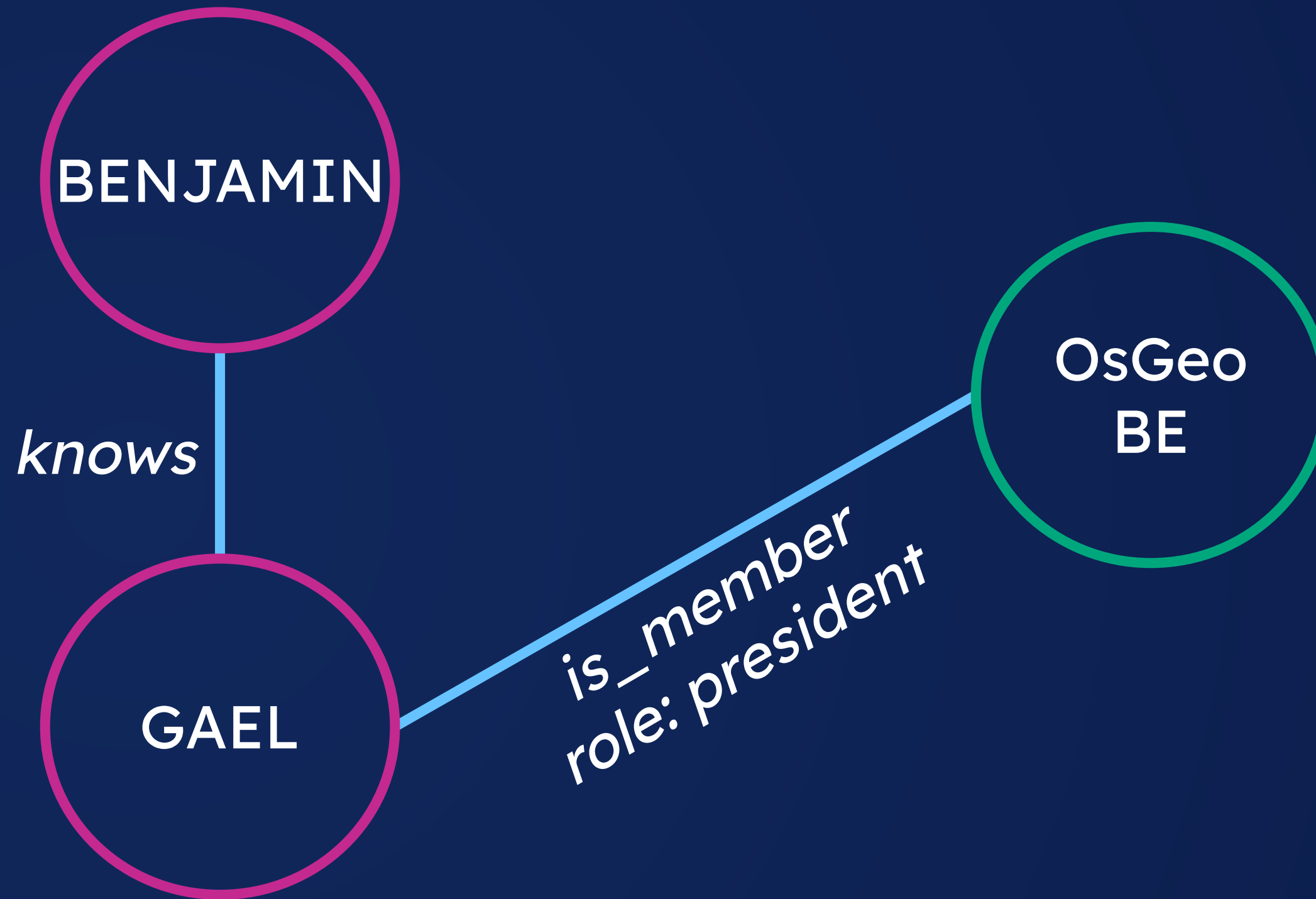
What is a graph?

Quick introduction



- **Nodes**: a, b, c
Person, POI, Device, ...
- **Edges**: X, Y
Friendship, Interaction, Route,...
- *Directed / Undirected graphs*
- *Properties / attributes*
- *Graph $\approx$ network*

Example



Exercise

Let's create new connections!



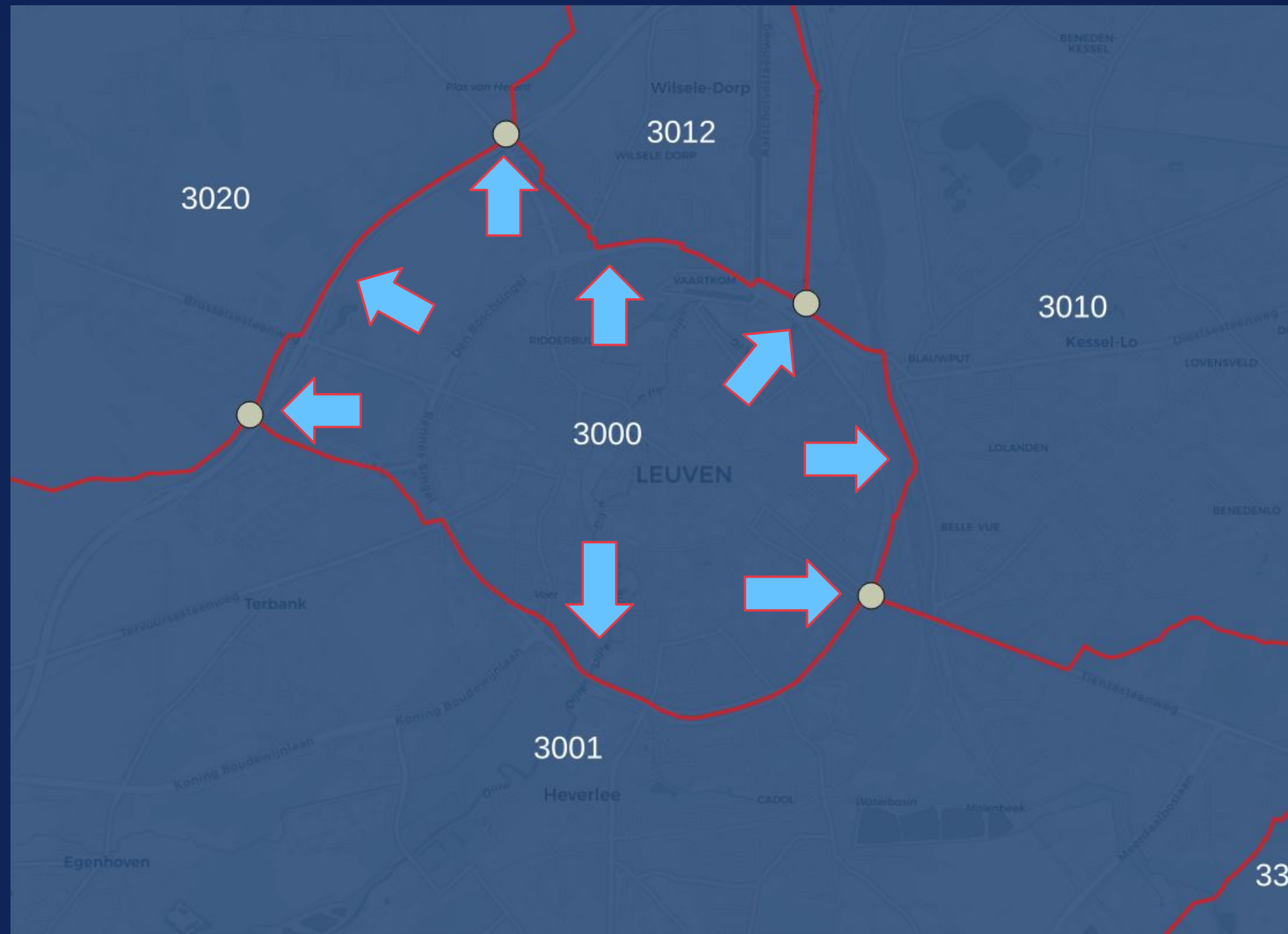
Graph-driven analysis

- **Centrality:**
degree, betweenness, closeness, PageRank
- **Community detection:**
Louvain, clustering, label propagation
- **Path & connectivity:**
*shortest path, connected components, max flow
/ min cut*
- **Graph mining:**
GNNs, DeepWalk

02

Handling topology

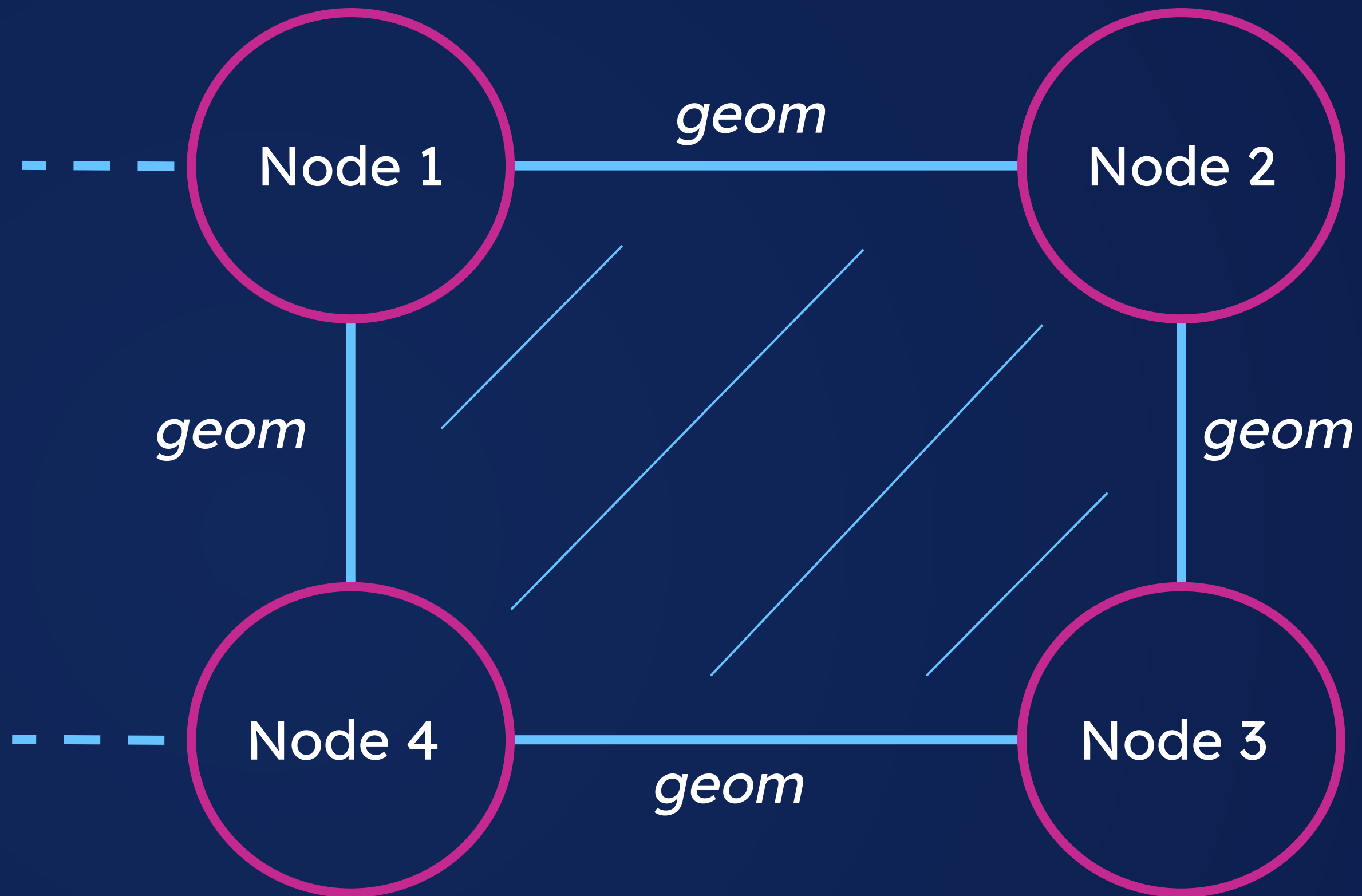
Postgis topology



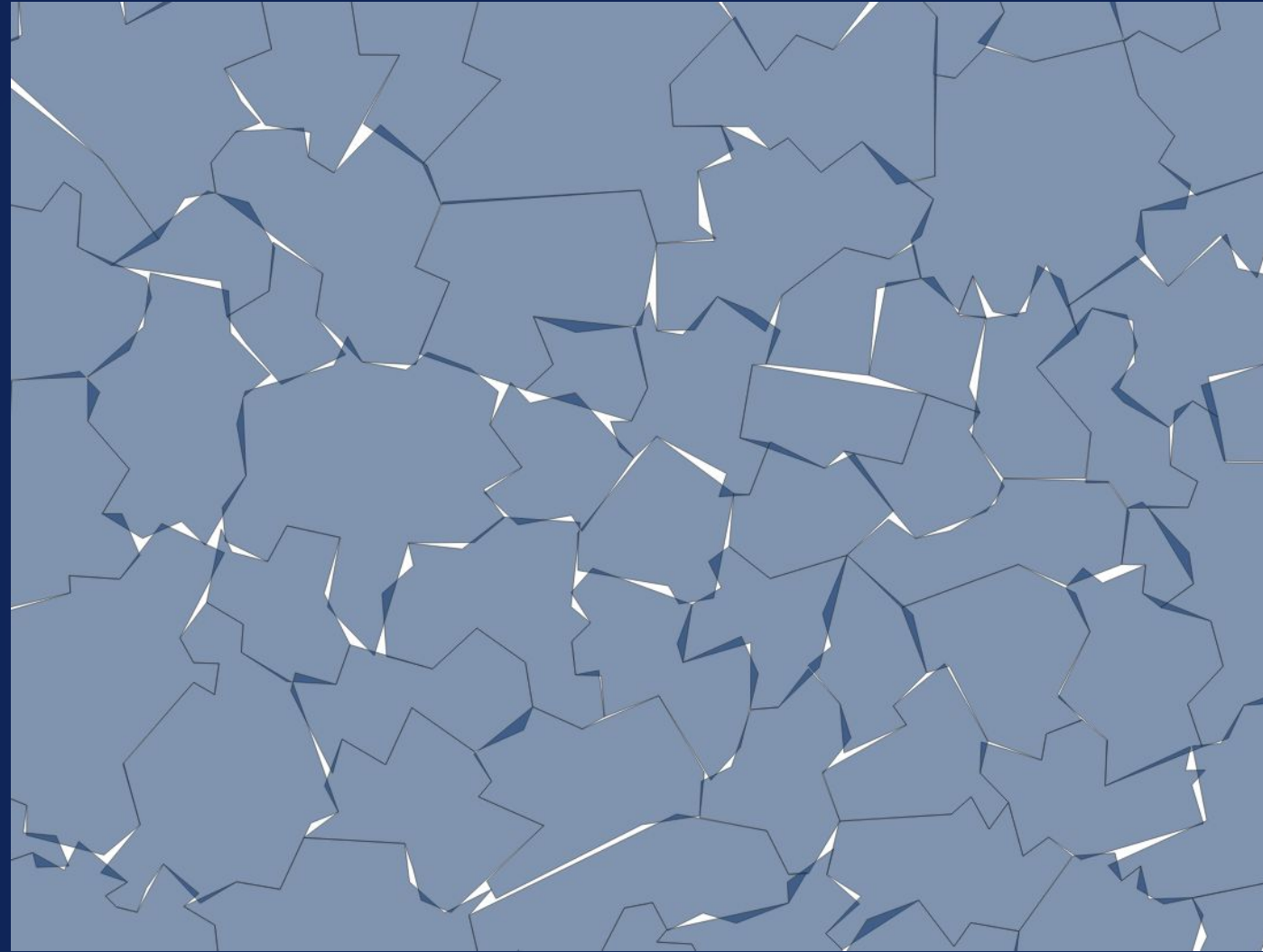
1 Polygon

- 4 Edges

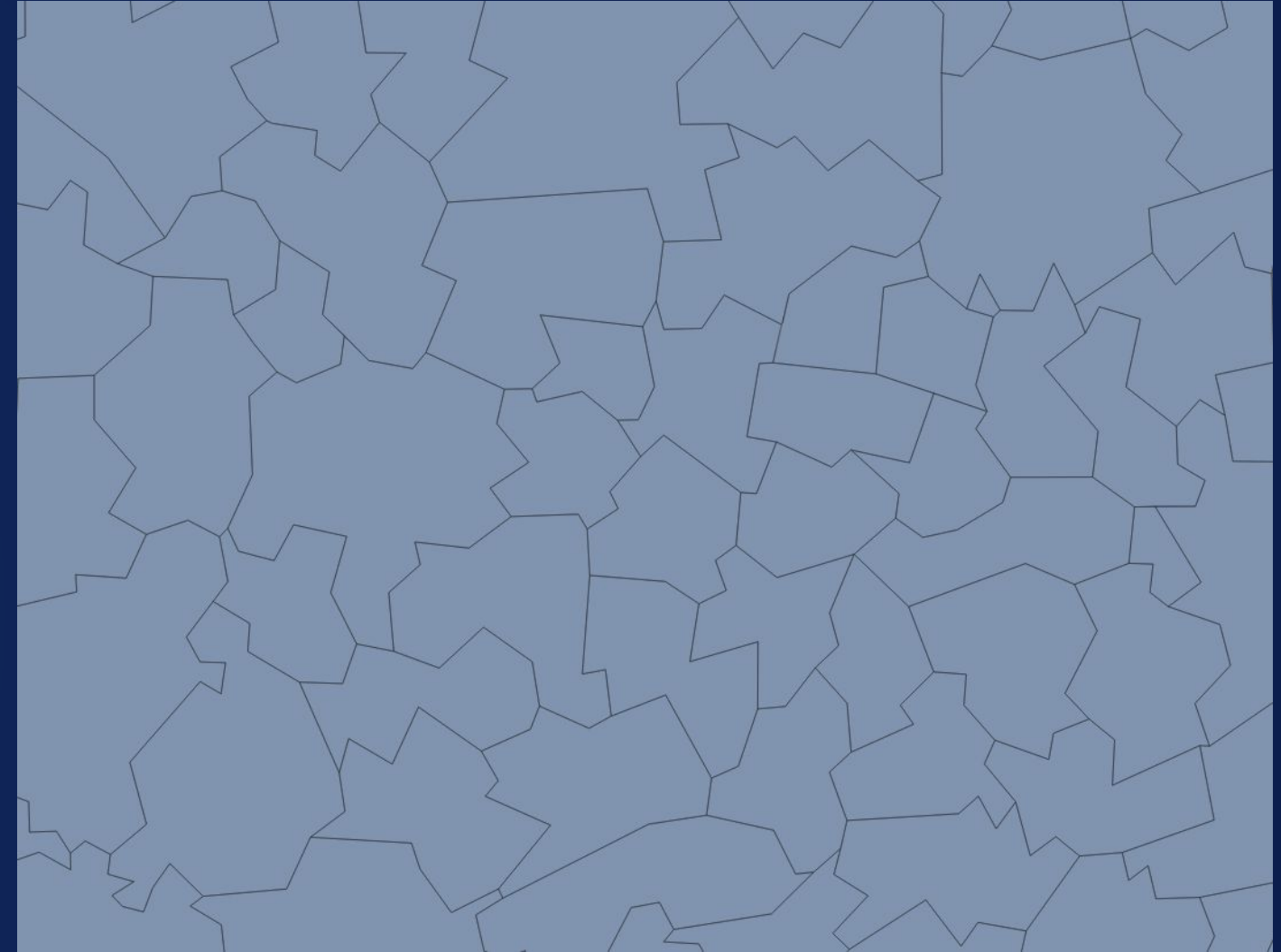
- 4 Nodes



Geometric simplification

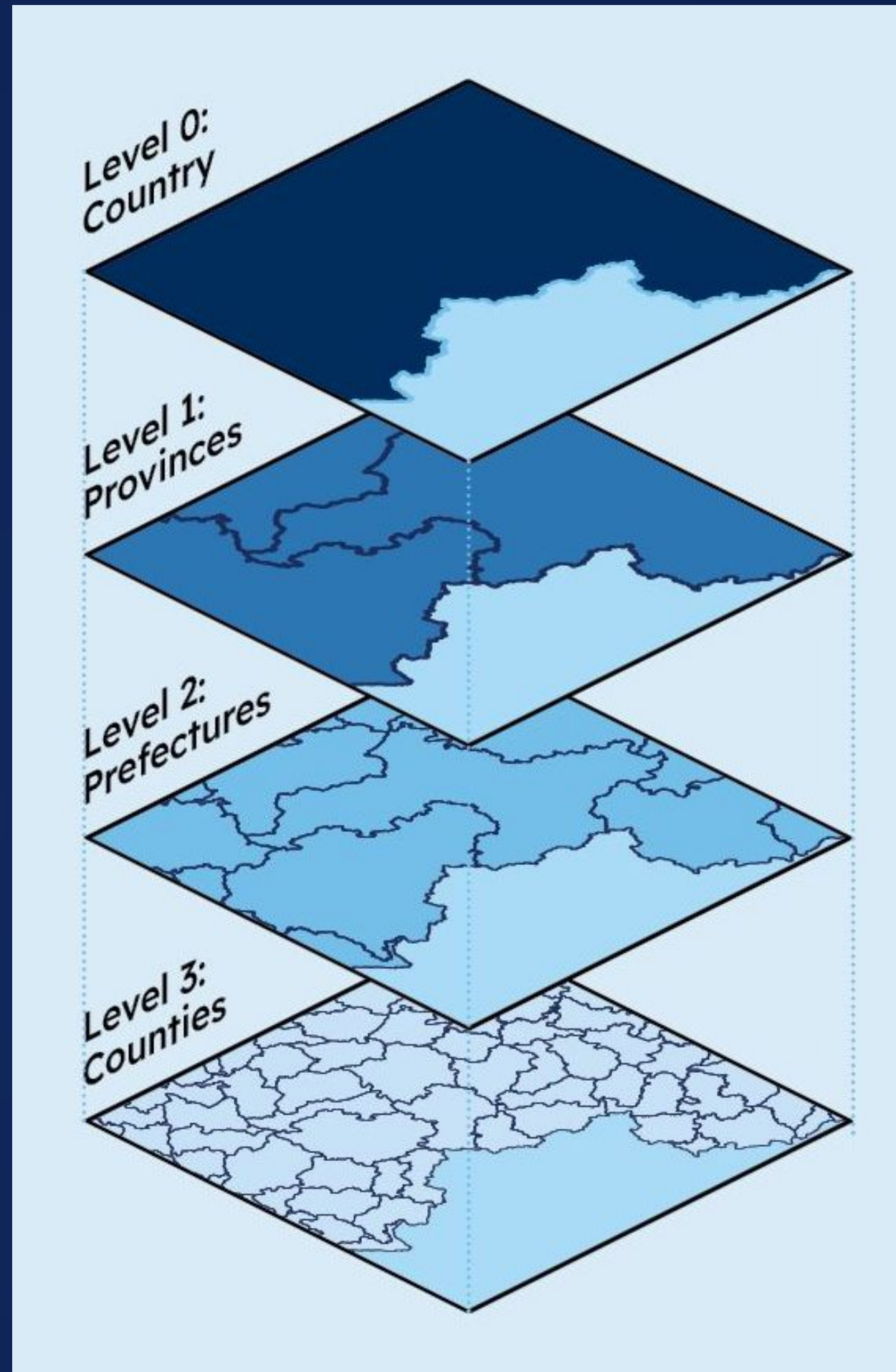


Without topological model



With topological model

Nested boundaries



*Layers “lasagna”
→ Nested boundaries*

Topology in postgis

- *Dedicated extension:* *postgis topology*
- *Query language:* *SQL*

03

Routing

pgRouting

What is Routing?



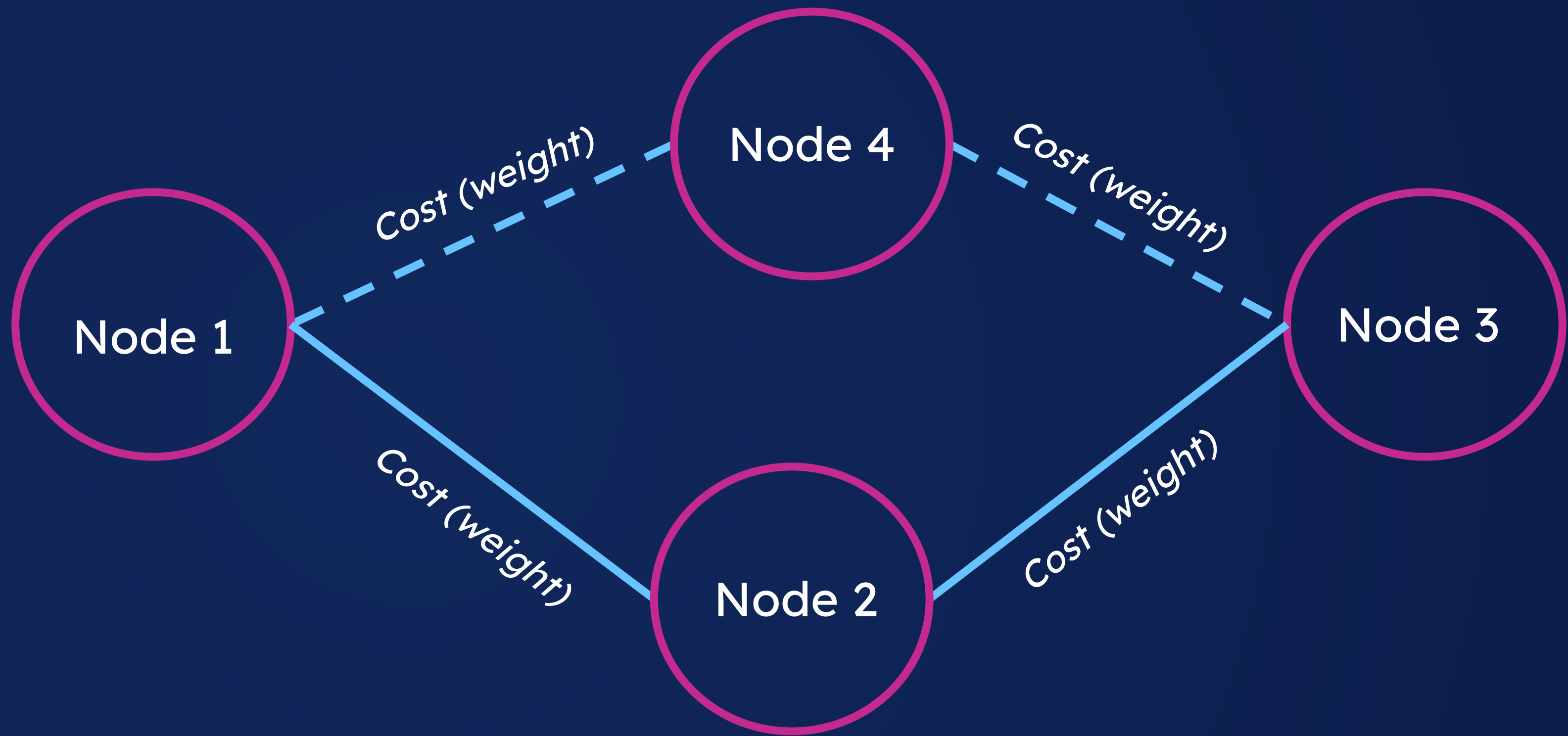
Shortest path between Mdina and Valletta (Malta)

Selecting paths in a network to go from a source to a destination.

→ Transform road geometries into graphs and apply algorithms to solve routing problems

Algorithms:

- *shortest path: Dijkstra, A**
- *Driving distances: reachable nodes within a cost limit (e.g. isochrones)*
- *Turn restrictions & costs: edge weights ÷ real world constraints*



Routing in postgis

- *Dedicated extension:* *pgRouting, MobilityDB*
- *Query language:* *SQL*



04

Knowledge graphs

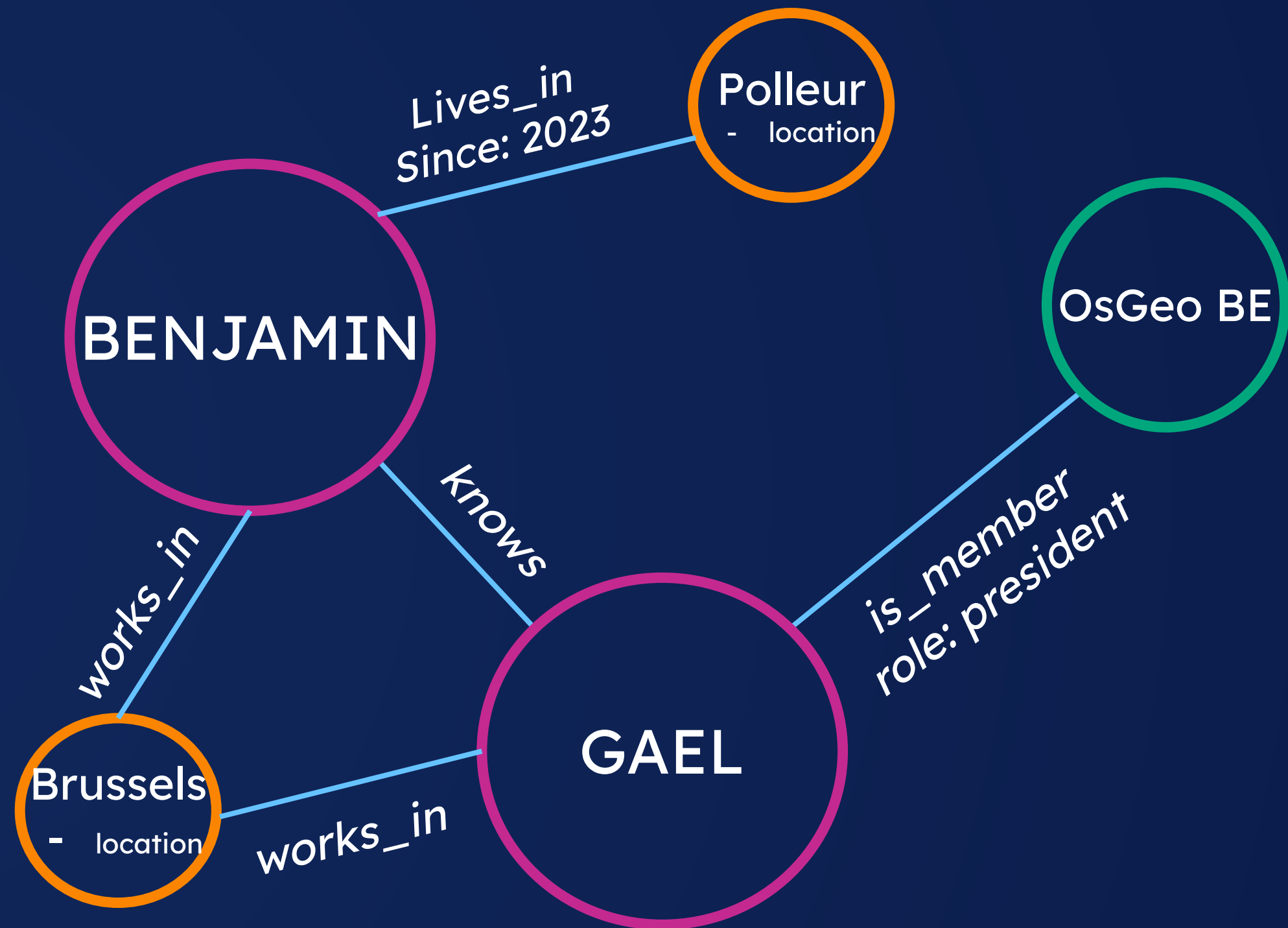
Neo4j

What is a Knowledge Graph?

Graph-based data model:

- Nodes (entities)
- Relationships (edges)

→ **Designed to capture connections, context and semantics.**



Neo4j graph database



- *Community edition = Open source*
- *NoSQL database*
- *Cypher query language:*

```
MATCH (p:Person) - [:WORKS_IN] -> (b:Place {name: "Brussels"})  
RETURN p.name AS PeopleWorkingInBrussels;
```

- *Optimized for interconnected data (index free adjacency), not just tables*
- *Graph data science toolset*
- *Supports some spatial data types and geospatial computations*

Why not do it in a relational database?

Feature	RDBMS (PostgreSQL)	Graph DB (Neo4j)
Data model	Tables, rows, columns	Nodes, edges
Relationship handling	FK, JOIN operations	Direct pointers between nodes (index-free adjacency)
Traversal perf	Multi-hop queries = multiple JOINS → can be slow	Multi-hop traversals are fast
Use case suitability	Tabular, flat, structured data	Connected, networked and semantic data
Flexibility	Schema changes are harder / rigid	Schema-free or flexible, easy to add new relationships

→ *Do we really have to choose?*

→ *No, you can combine them in a **polystore** infrastructure*

Applications in Geospatial Analysis

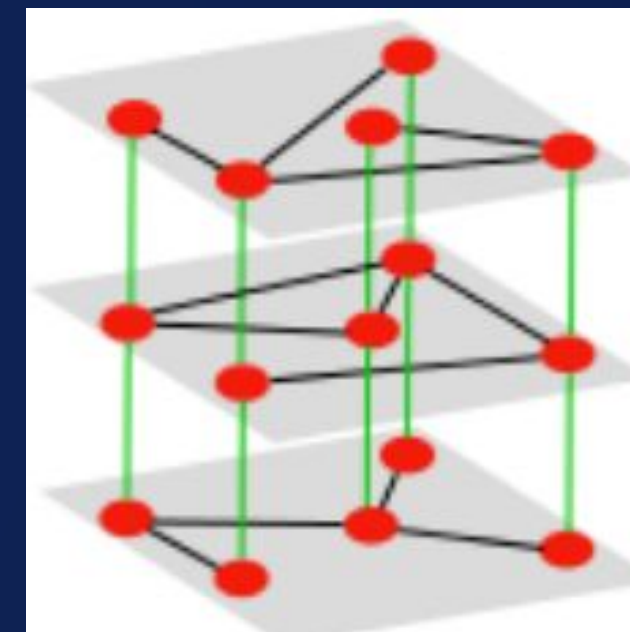


- *Interconnected features*
- *Spatial relationships*
- *Polystore:*
 - *RDBMS (postgis)*
 - *graph db (neo4j) becomes a tool*

Applications in Geospatial Analysis (2)



- *Interconnected features:*
 - *Geometrically VS by road*
 - *Different types of features*
- *Multi-layer graph (multiplexing)*



The future is bright!

Key takeaways:

- *GIS open source ecosystem already has great graph applications*
- *Graph & geospatial analysis work great together*
- *Knowledge graphs can enrich your geospatial analysis:
where things are but also how they are related in space*

Open perspectives (bonus): QGIS plugin for Neo4j